



Problem solving with place value

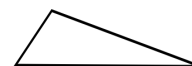
Task A: Same measure, different units

Here is an important reminder about two types of triangles.

An isosceles triangle has two sides which are the same length.



A scalene triangle has no sides with the same length.



For each set of measurements, deduce if they are the lengths of a scalene or an isosceles triangle.

1) 1.739 m, 173.9 cm, 1793 mm

E.g. Only the lengths 1.739 m and 173.9 cm have the same digits in the same order. These are the lengths I need to check.

$$1.739 \text{ m} \times 100 = 173.9 \text{ cm}$$

These lengths are the same. This is a set of lengths for an isosceles triangle.

2) 0.032 km, 3020 cm, 32 000 mm

E.g. Only the lengths 0.032 km and 32 000 mm have the same digits in the same order. These are the lengths I need to check.

$$0.032 \text{ km} \times 1000 \times 100 \times 10 = 32\,000 \text{ mm}$$

These lengths are the same. This is a set of lengths for an isosceles triangle.

3) 5.1 cm, 0.000053 km, 50.3 mm

E.g. None of the lengths have the same digits in the same order. None of the lengths will be the same when converted.

None of the lengths are the same. This is a set of lengths for a scalene triangle.

4) Izzy has measured the sides of a triangle.

Why might you think that Izzy has made an error?

E.g. Izzy has measured one length in mm. This might be ok but when converted to cm, it is 0.93cm. This seems a lot smaller than the other measurements.



I measured lengths of 31.2 cm, 24.9 cm and 9.3 mm

How could you check? *E.g. I could remeasure that side to see if I get the same measurement again. I could try to draw a triangle with Izzy's measurements.*



Task B: Determining useful units of measure

1) Fill in the gaps using an appropriate unit of measure from the box.

I recently visited Paris. It took 2 *hours* and 37 *minutes* to get there by train. I visited the Eiffel Tower, which is 330 *metres* tall. Whilst at the Eiffel Tower, I visited a souvenir shop and bought 0.5 *kilograms* of sweets to share with my friends.

2) Which metric measures might be used to measure the following?

- a) Length of an ant *millimeters*
- b) Mass of a person *kilograms*
- c) Capacity of a bucket *litres*
- d) Height of a house *metres*

3) Which imperial measures might be used to measure the following?

- a) Length of an ant *inches*
- b) Mass of a person *stones or pounds*
- c) Capacity of a bucket *pints*
- d) Height of a house *feet*